

Date: 2026-08-05

Request ID: 100048809

Hotplate		
Analyte(s)	Ag	
SOP#	Determination of % Silver Composition	
Cocktail		
sample(s)	weight (g)	volume (mL)
200128703_1	0.5019	200
200128703_2	0.5051	200

Hotplate		
Analyte(s)	Pd	
SOP#	Determination of Palladium in Catalyst with or without Ag	
Cocktail		
sample(s)	weight (g)	volume (mL)
200128703_1	1.0055	200
200128703_2	1.0056	200

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128703	16.5627	18.5836	18.5697	0.69	0.9931

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128703_1	10/9	0.804	358.45	0.04
200128703_2	10/9	0.793	351.30	0.04

ag result: 354.87 ppm

ag result: 0.04 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Pd ICP analysis

sample	Dilution	conc. [mg/L]	Pd_ppm [dried]	%Pd [dried]
200128703_1	10/9	1.918	426.83	0.04
200128703_2	10/9	2.107	468.84	0.05

pd result: 447.83 ppm

pd result: 0.04 %

pd lot: *Manufacturer: Ricca, 841005E exp: 2027-11-30*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$